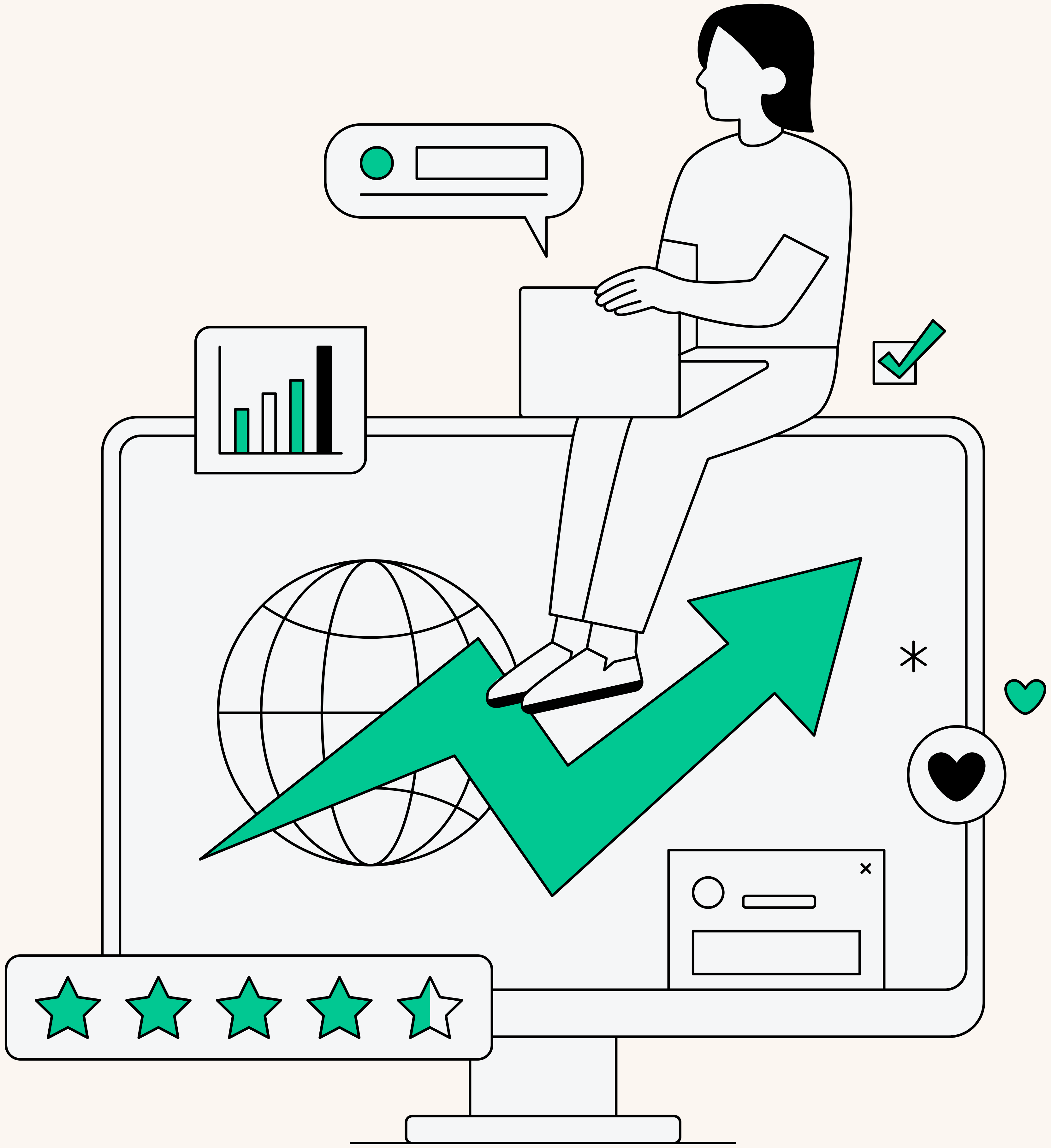
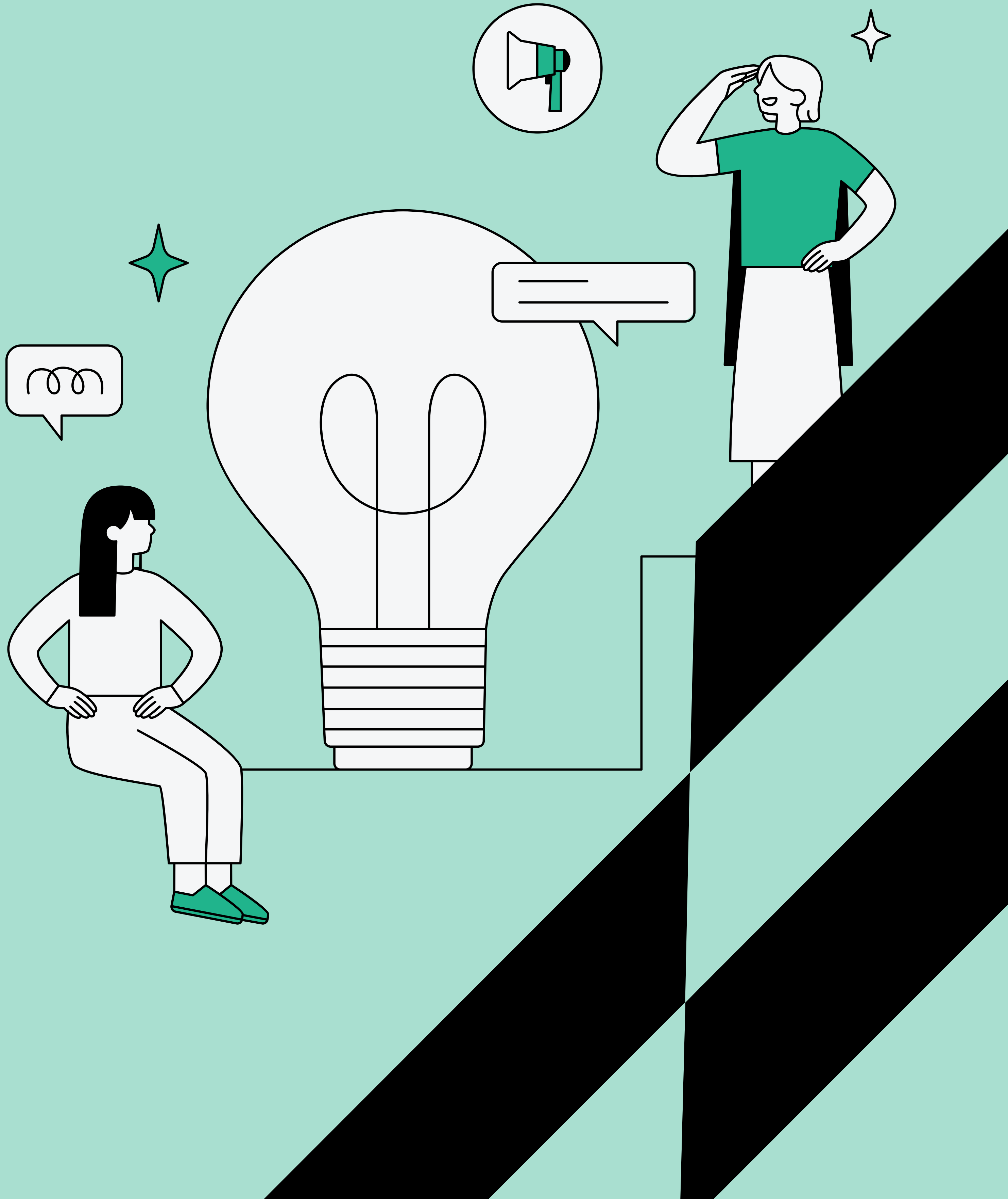


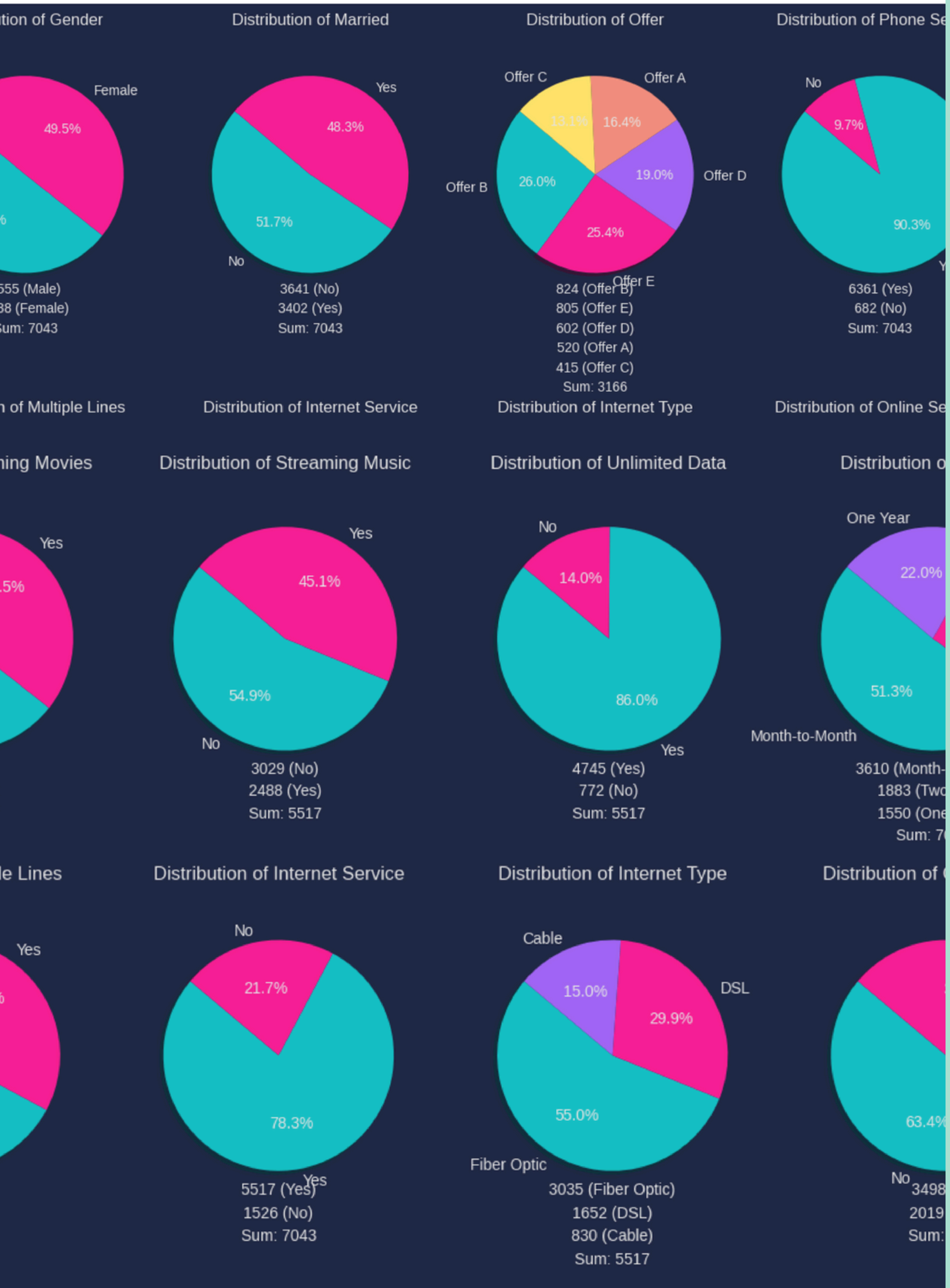
TELECOM CHURN PREDICTION

Presented by: M. Kirubha

EXECUTIVE SUMMARY

- This report provides a comprehensive analysis of customer churn within the telecom sector, highlighting key drivers of churn and predicting potential customer segments most at risk. By leveraging advanced machine learning techniques, the analysis identifies critical areas where targeted retention strategies can significantly reduce churn.
- Our findings indicate that a 5% reduction in churn could lead to a projected revenue increase of $1.2M annually, offering substantial financial impact. This presentation focuses on data-driven recommendations to optimize retention strategies and increase customer lifetime value.





This analysis utilizes these attributes to develop a predictive model that provides actionable insights for customer retention.

DATA OVERVIEW

The dataset comprises 7,043 customers, with variables that capture customer demographics, service usage, and account tenure. Key features include:

- Contract Type: Month-to-month customers show a higher churn tendency, providing an opportunity to target them with loyalty programs.
- Monthly Charges: Higher monthly charges are correlated with increased churn, suggesting that customers with premium services may need more value-focused offerings.
- Tenure: Long-term customers exhibit higher loyalty, making tenure a key feature in predicting churn likelihood.

KEY INSIGHTS FROM DATA ANALYSIS

1 Contract Type

Customers with month-to-month contracts are 7.26 times more likely to churn than those with longer-term contracts (one or two years).

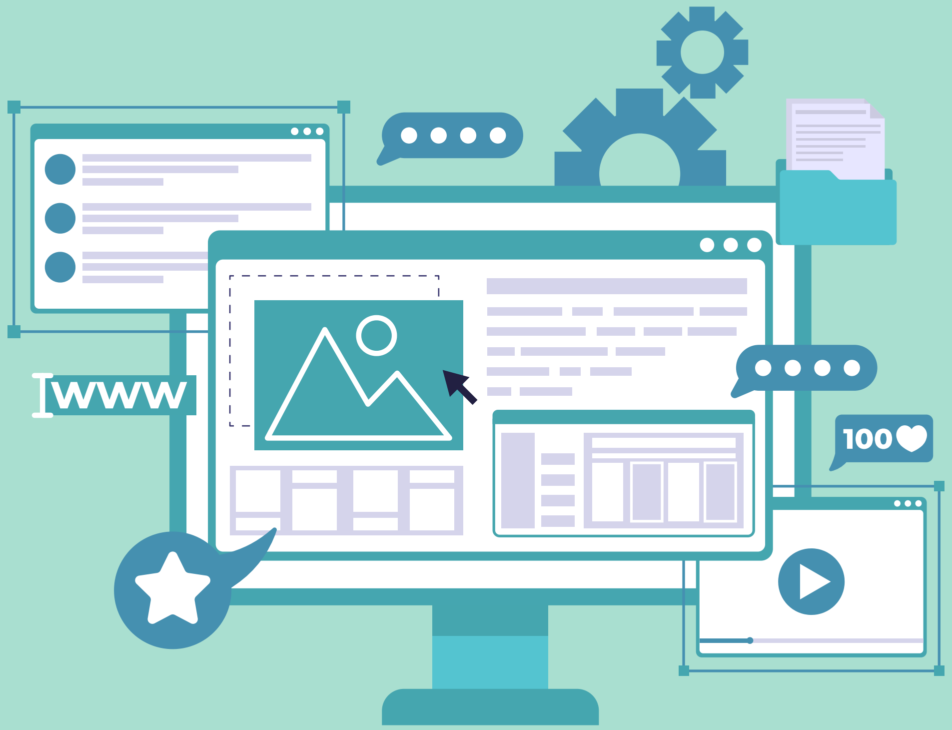
2 Monthly Charges

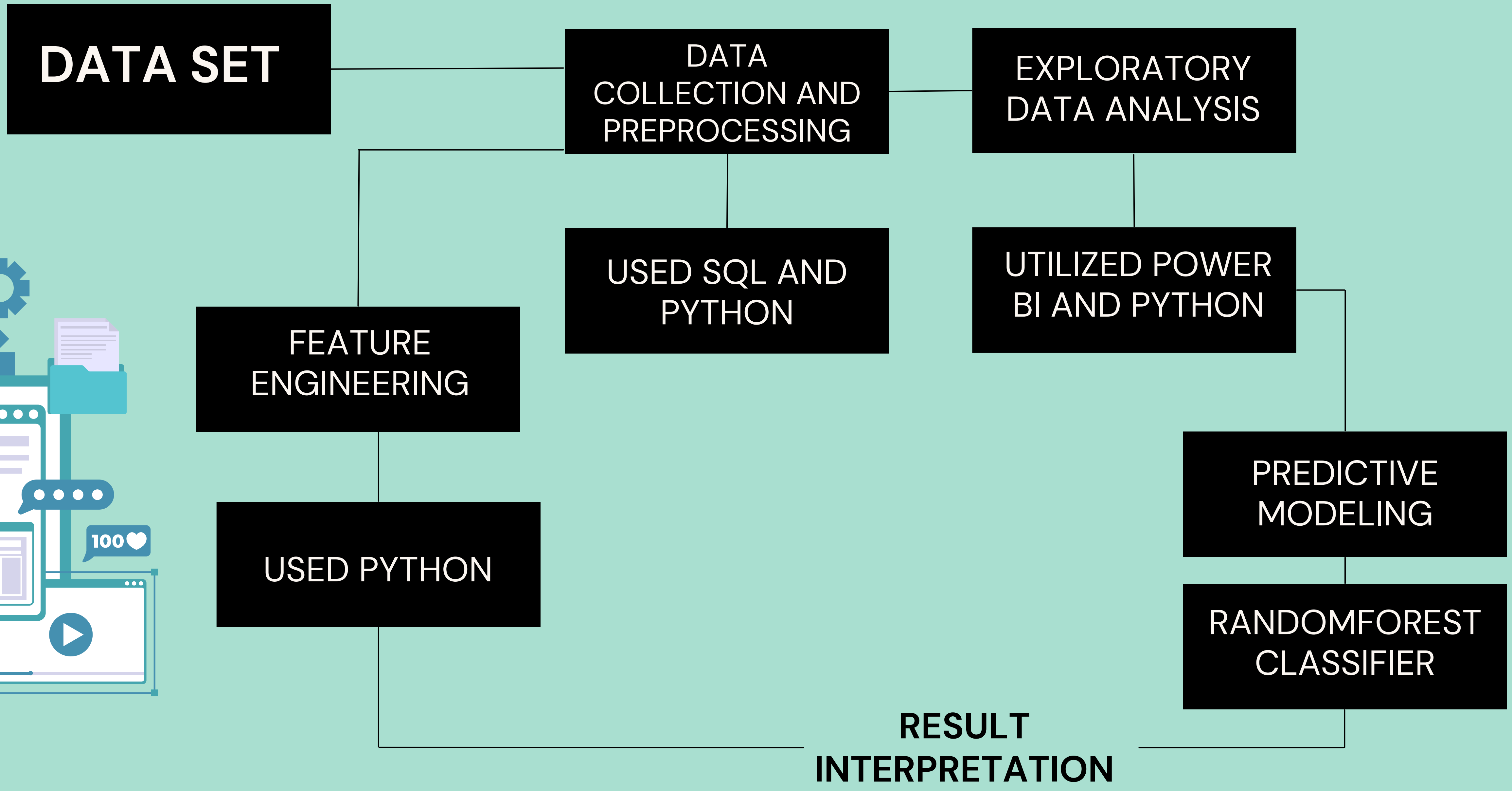
Customers with monthly charges in the range of $68.8 to $106.75 are 2.64 times more likely to churn.

3 Service Usage

Customers who subscribe to premium services, such as streaming TV or online security, tend to have significantly lower churn rates.

METHODOLOGY





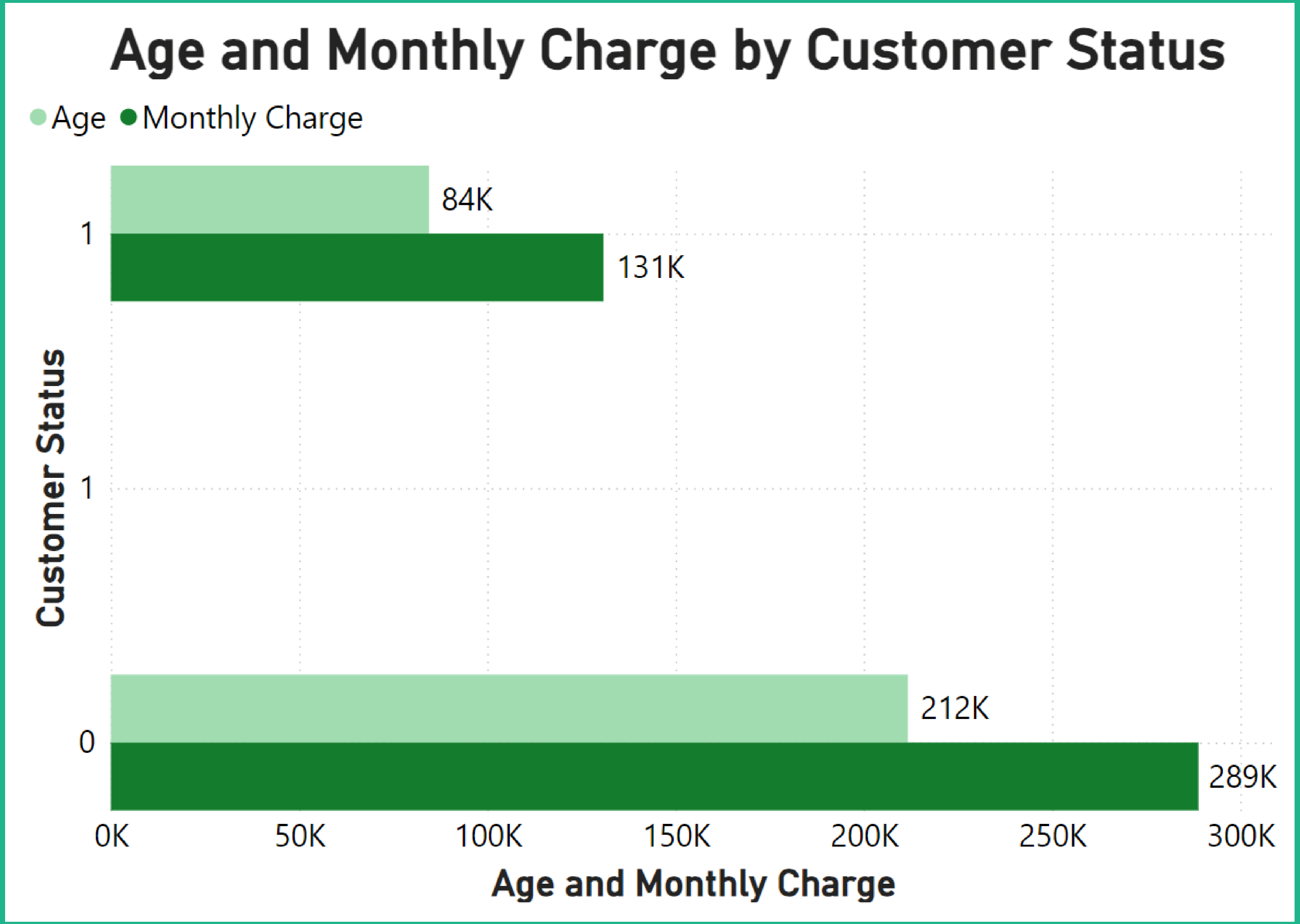
DEEP DIVE INTO CUSTOMER SEGMENTATION

High-Risk Segments

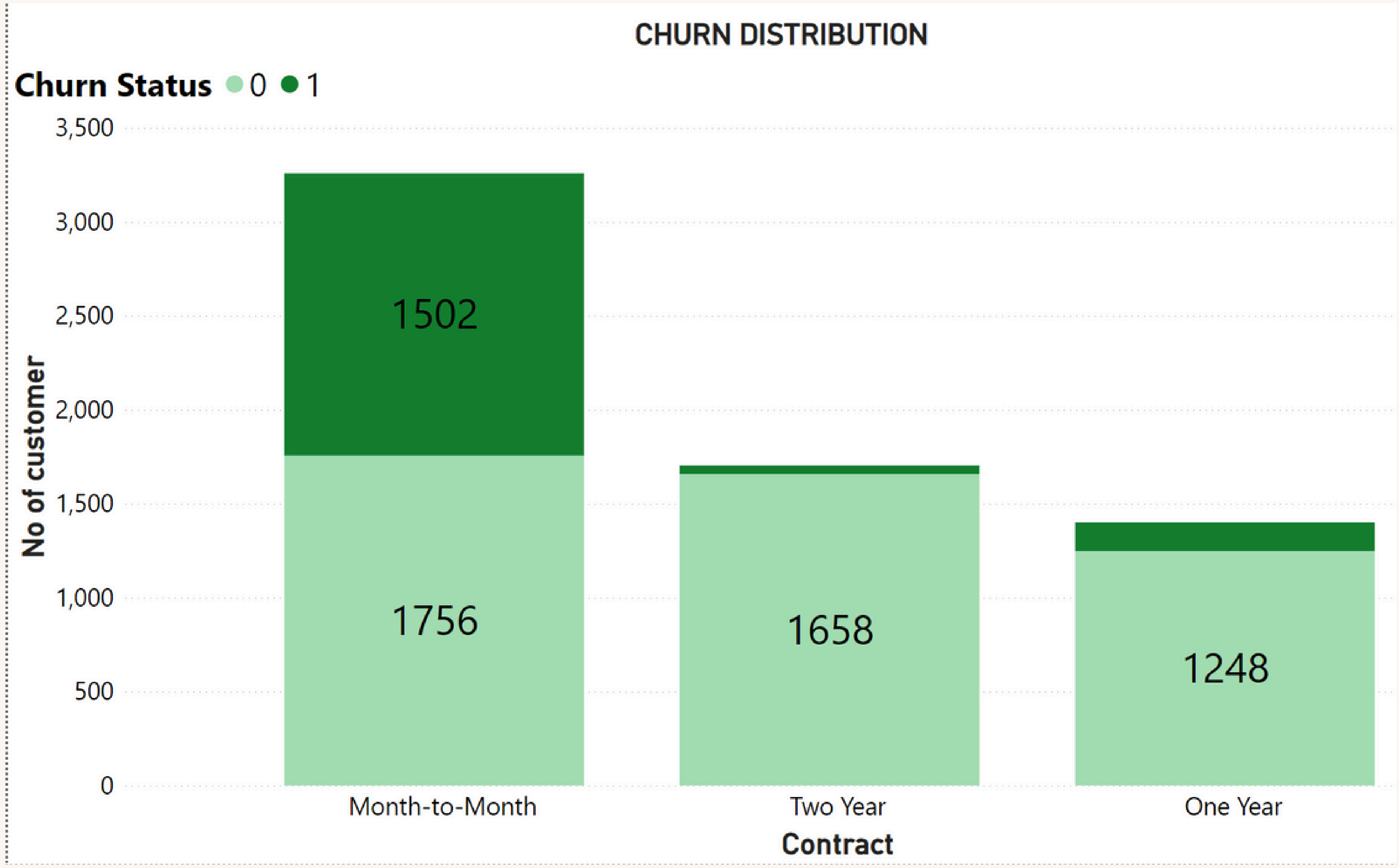
- Customers aged 25-40, especially those with month-to-month contracts and high monthly charges, are at the highest risk of churning.
- Younger customers (under 30) are more price-sensitive and churn at higher rates compared to older customers.

Loyalty Drivers

Customers aged 50+ with long-term contracts tend to show greater loyalty and are less likely to churn.

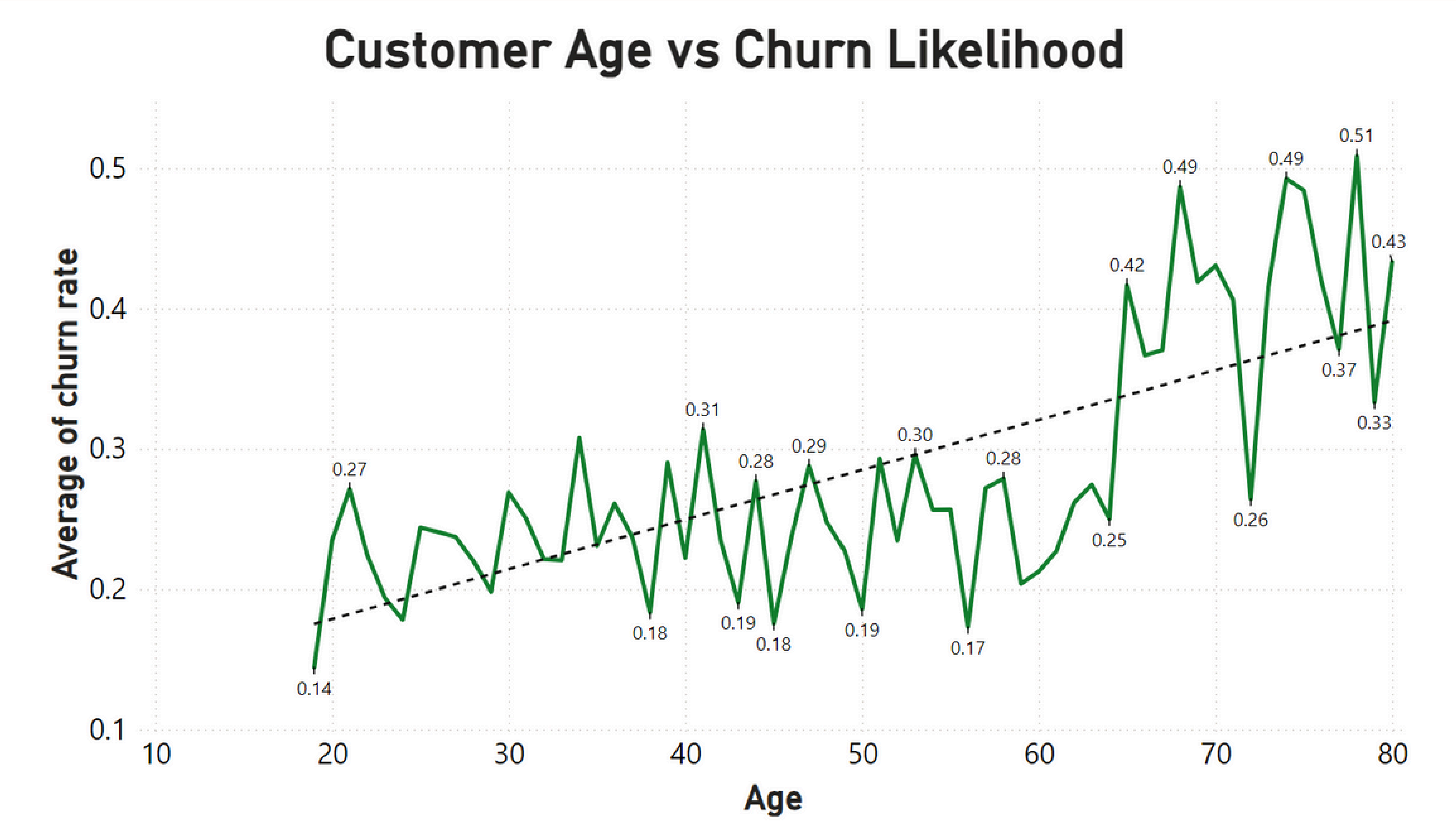


CHURN RATE BY CONTRACT AND CHARGES



MONTHLY CONTRACT CHURN:

47% of customers with monthly contracts churned, compared to only 11% for those on long-term contracts.



CHURN RATE BY CUSTOMER AGE

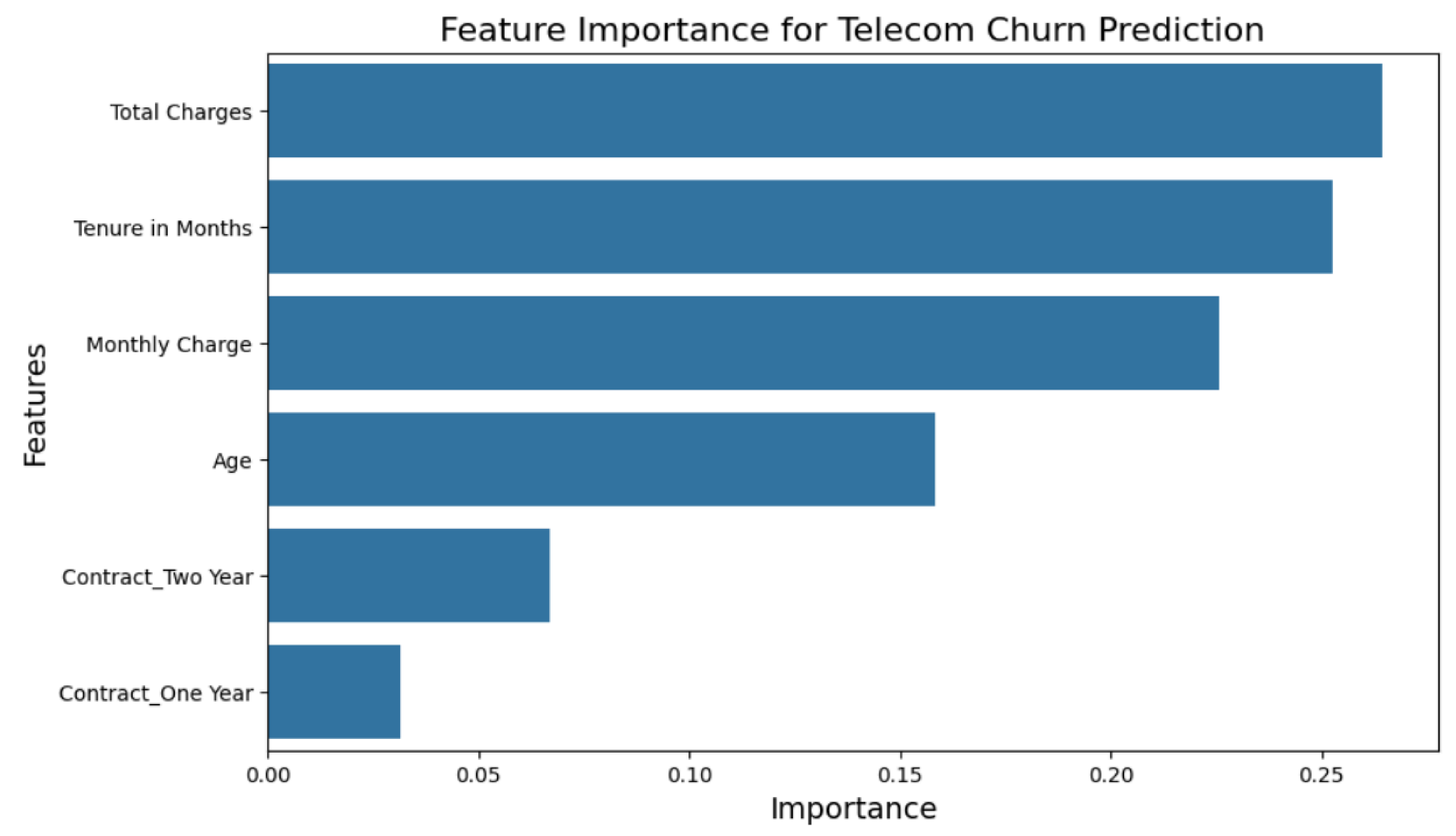
CHURN LIKELIHOOD INCREASES WITH AGE:

The chart shows that as customer age increases, the likelihood of churn rises, with churn rates reaching over 50% for customers in their 70s and 80s.

VARIABILITY IN MIDDLE AGE GROUPS

Middle-aged customers (30-50) exhibit fluctuations in churn, ranging from 18% to 31%, suggesting other factors may influence churn in this group.

Key Factors Influencing Telecom Churn

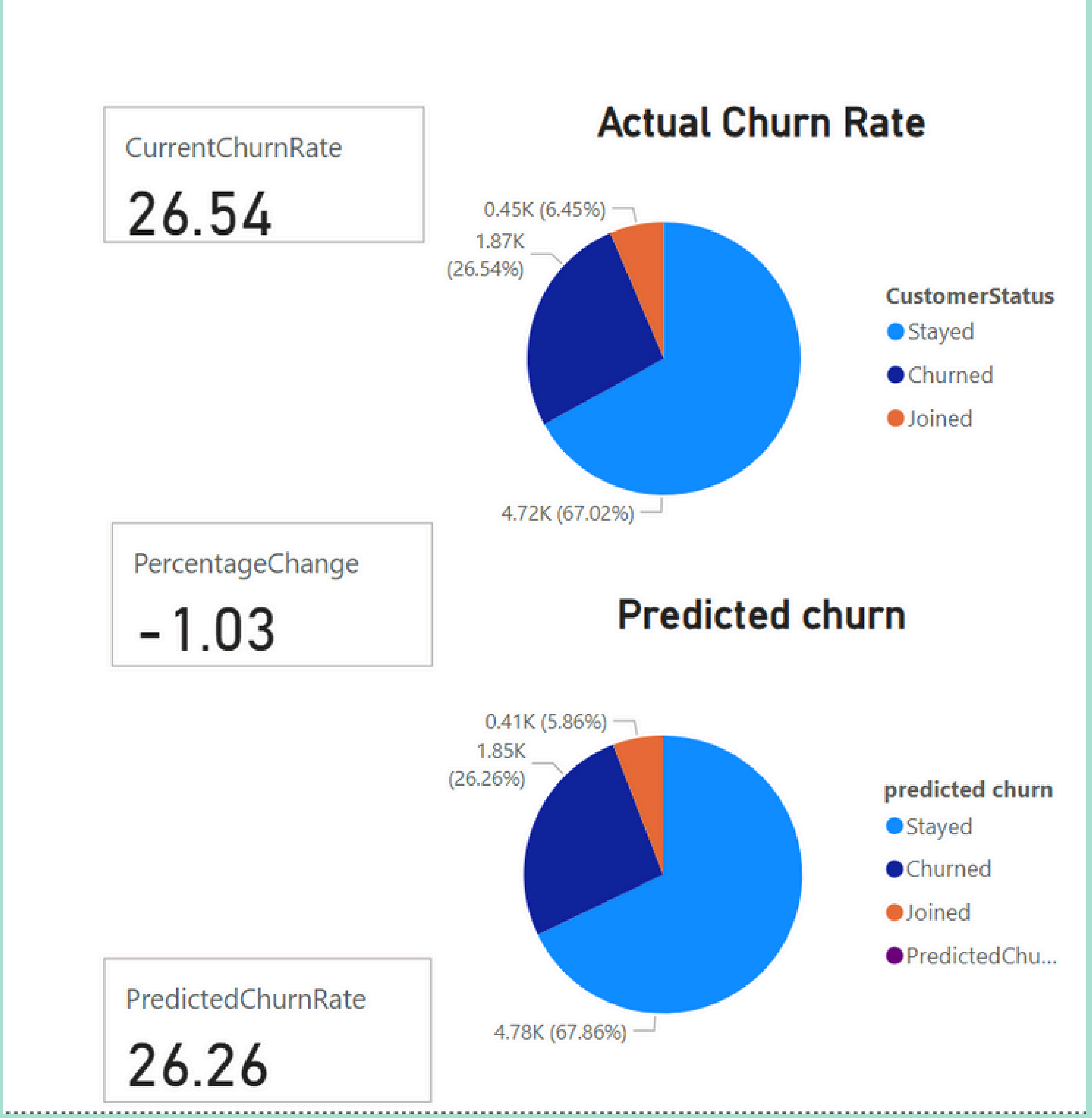


Feature Importance

- Total Charges and Tenure are the top predictors of churn.
- Higher monthly charges correlate with increased churn likelihood.
- Customers on longer contracts tend to be more stable but still require engagement to retain them.
- Age can influence churn, with older customers at higher risk.

Predictive Modeling Results

Key features influencing churn include contract type, monthly charges, and tenure. These results enable targeted marketing interventions for high-risk customer segments..

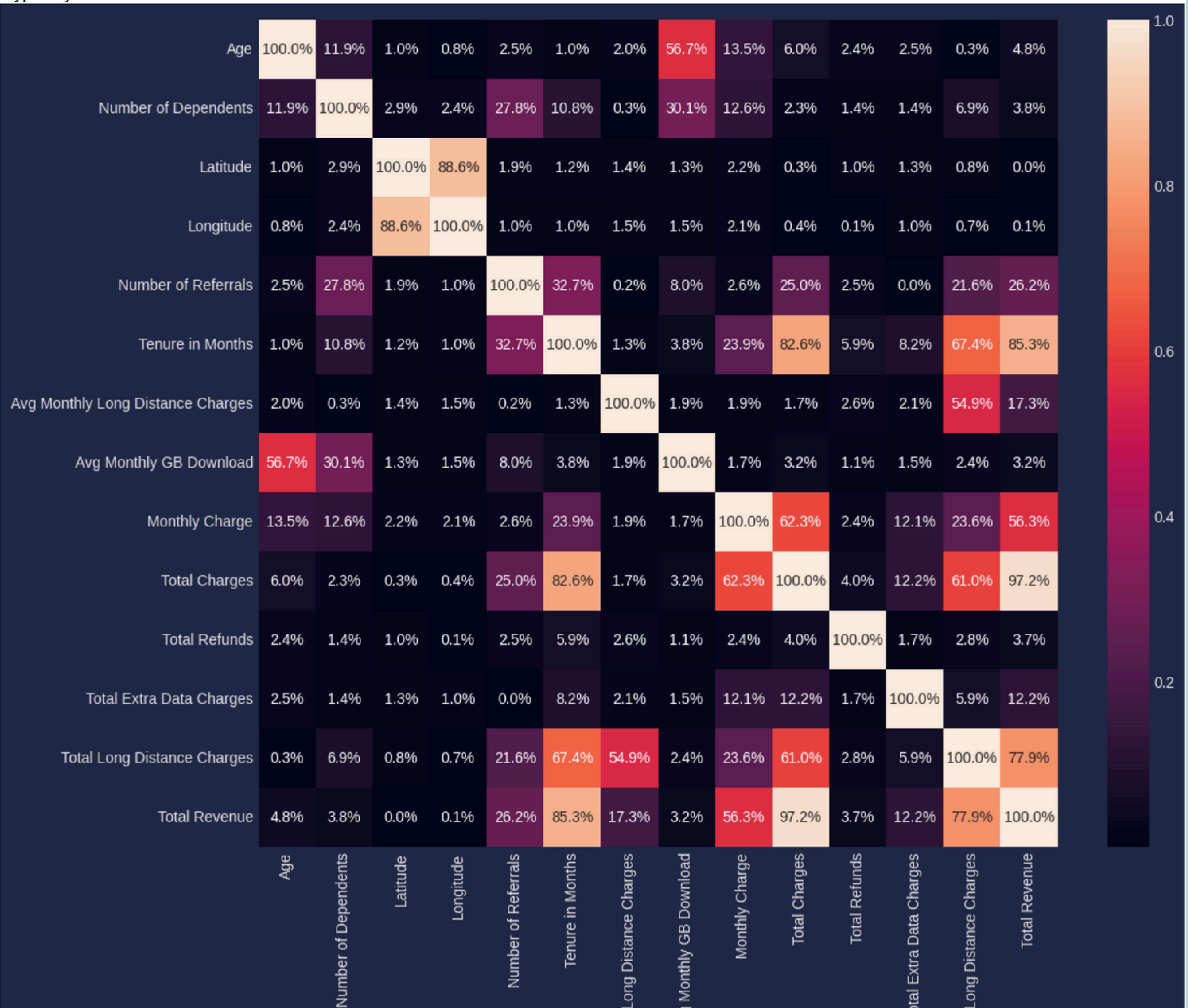


A RandomForestClassifier was trained on customer data to predict churn.

Accuracy: After evaluating several models, including Logistic Regression, Decision Trees, and Random Forest, the RandomForestClassifier was selected based on its superior performance, achieving an accuracy of 83% with an F1 score of 0.78

Important Features:

Churn Prediction: The model identified that 23% of current customers are likely to churn within the next 6 months.



CORRELATION OBSERVATION

MONTHLY CHARGES CORRELATION

Customers with charges between $70 - $100 are at the highest risk of churn.

Recommendations

Focus on Month-to-Month Customers

- With a churn rate exceeding 45%, customers on month-to-month contracts should be the primary focus of retention efforts. Offering incentives such as discounts or upgrading them to annual contracts with added benefits can reduce churn in this group.

Address High-Paying Customers

- Customers with higher monthly charges exhibit a churn rate of 35%, indicating dissatisfaction with premium services. Introducing loyalty programs or bundling value-added services at a discounted rate can improve retention among these high-revenue customers.

Target Older Age Groups

- With churn rates peaking at 50% for customers over 60, personalized plans or pricing adjustments for senior customers may help retain this valuable segment.

Marketing Strategies

Customer Retention Campaigns

- Offer discounted long-term contracts to month-to-month customers to incentivize longer commitments.
- Provide personalized discounts or bundle offers to customers with high monthly charges.

Premium Services Upsell

- Encourage more customers to subscribe to premium services like streaming and online security, which are correlated with lower churn.

Target Younger Customers

- Develop targeted retention campaigns for younger customers (under 40), with price-sensitive offers and flexible contract terms.

POTENTIAL REVENUE IMPACT

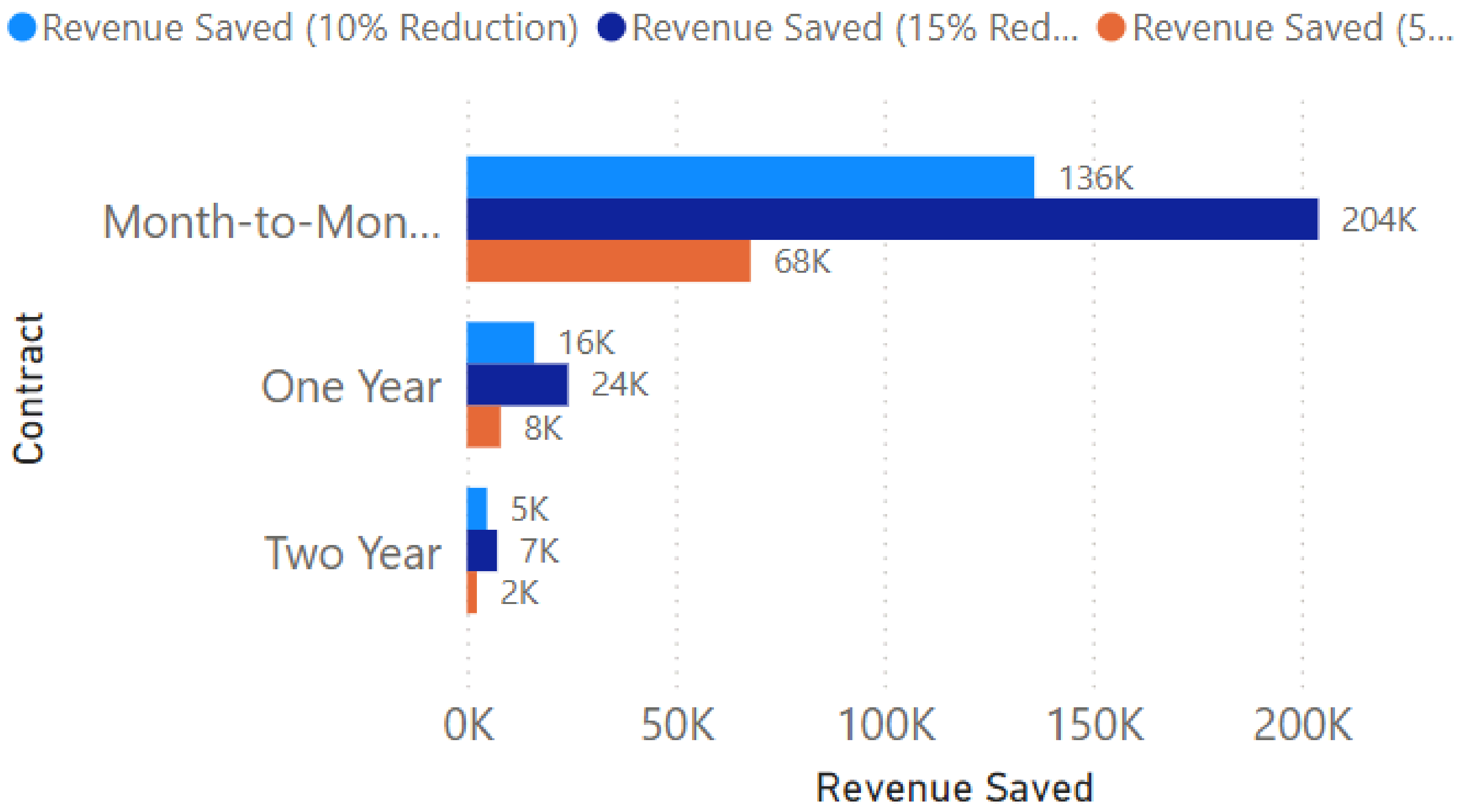
- A projected 5% reduction in churn could lead to a revenue increase of $1.2M annually. This estimate is based on the current churn rate, customer lifetime value (CLV), and the average monthly revenue per customer.

- The calculation assumes that reducing churn among high-risk segments (e.g., month-to-month contracts, high-paying customers) will significantly extend the customer lifespan and increase total revenue.

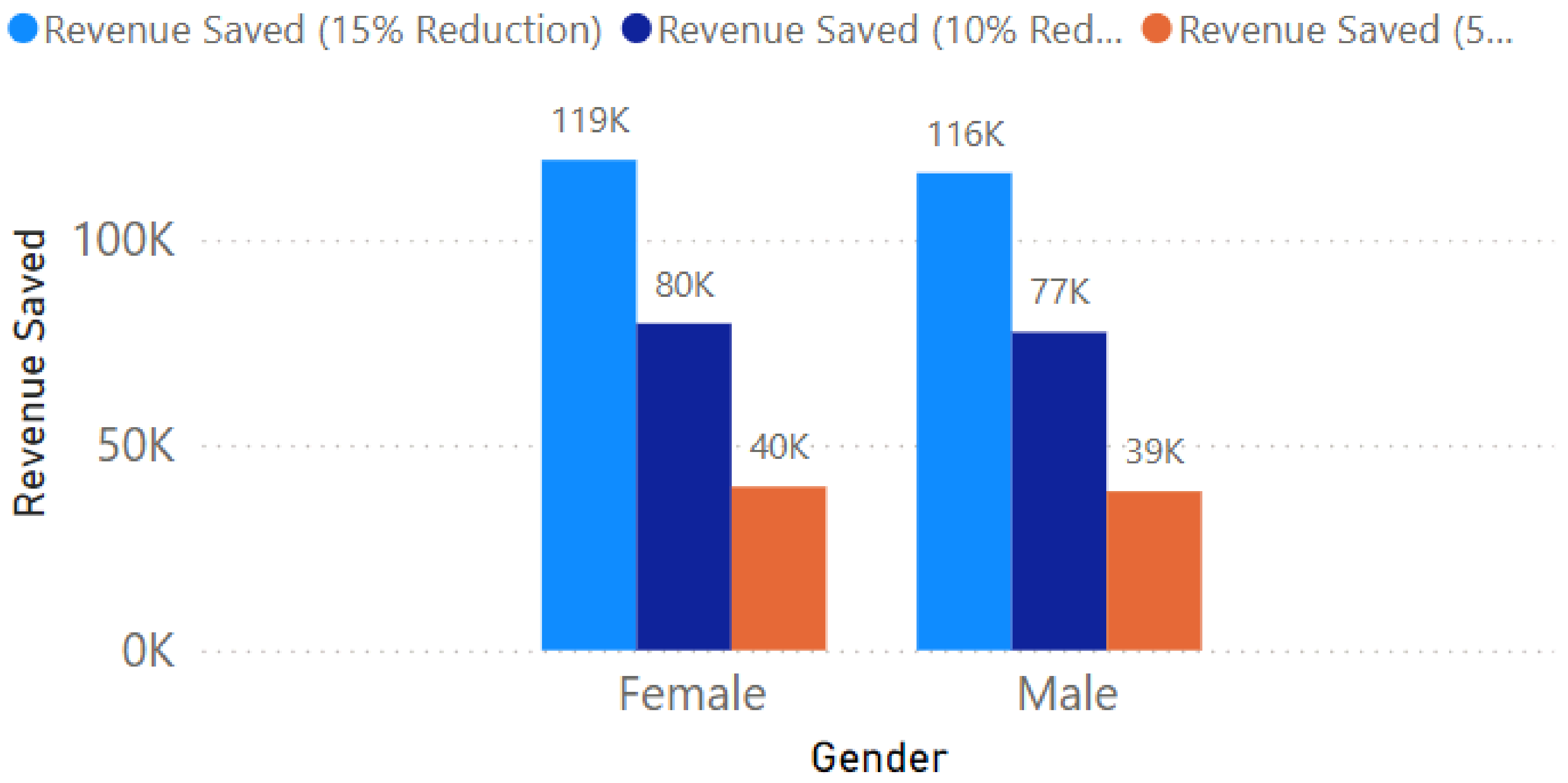
- This financial impact underscores the importance of targeted retention strategies for maximizing long-term profitability.

Financial Impact of Churn Reduction-Revenue Saved Analysis

Revenue Saved by Contract



Revenue Saved by Gender



REVENUE LOST

1.57M

MonthlyChargeCategory

High | Low | Medium

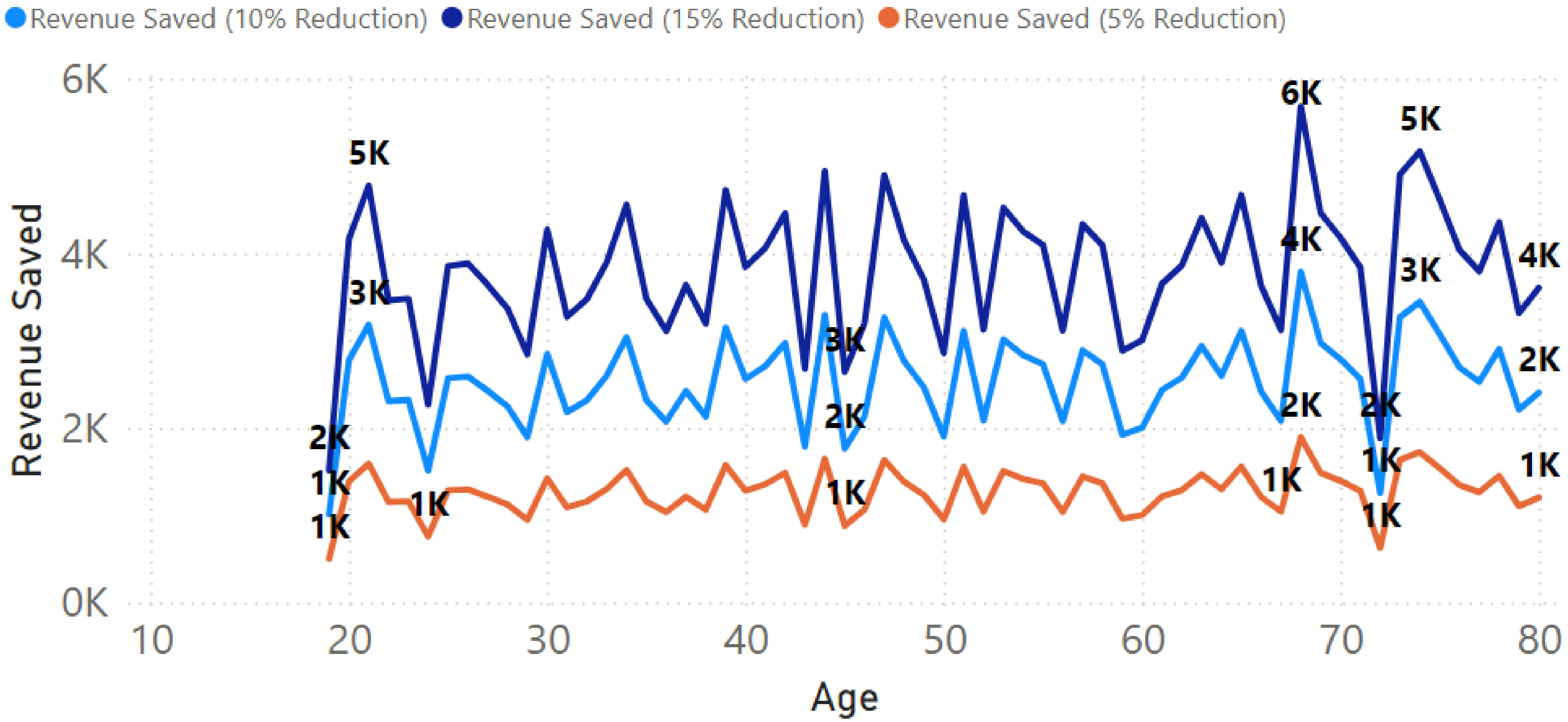
Age (bins)

26

Revenue Saved Through Churn Reduction

REVENUE SAVED(5%)	REVENUE SAVED(10%)	REVENUE SAVED(15%)
78.52K	157.03K	235.55K

Revenue Saved by Age



THANK YOU

Conclusion

By identifying at-risk customers and implementing targeted retention strategies, the company can significantly reduce churn and boost revenue. The predictive model provides actionable insights into customer behavior, and the recommended strategies are designed to reduce churn by focusing on the most vulnerable segments of the customer base.